

Design of microgestures for human–robot collaboration scenarios

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Abstract—In application scenarios with extreme environments, spatial constraints, and high fatigue loads, traditional large-limb gesture interactions involving wrists, shoulders, and elbows are not applicable, and an interaction method with low motion and low fatigue is needed as an alternative. In this study, a user-inspired approach is used to explore heuristic microgestures for human–robot interaction scenarios in a desert environment, where an operator operates an intelligent rover at a medium distance to accomplish tasks such as azimuthal movement and derives an optimal vocabulary of microgestures suitable for typical tasks. This interaction method controls the robot through tiny finger movements, which are flexible, reduce the user's physical exertion, enable it to maintain good performance during long working hours, and provide a natural and efficient interaction experience.

Keywords—human–robot collaboration, microgesture, gesture design, user-inspired methods, human–robot interaction

I. INTRODUCTION

In today's era of rapid technological development, human–robot interaction has become a key link in many fields [1]. Some intelligent robots integrate artificial intelligence, computer science and other multidisciplinary technologies [2], which, to a certain extent, satisfy some people's needs. However, owing to the influence of various factors, such as the interaction interface, interaction equipment, interaction modality and interaction mode, a series of urgent problems, such as the user experience [3], task efficiency [4], and system reliability [5], have emerged in human–robot interactions at various levels. Moreover, in the face of complex operation tasks, users need much time and energy to learn and adapt to the interaction mode, which leads to low efficiency in task completion; the unsmoothness of the interaction also affects the user experience and reduces the user's satisfaction with the system. Some studies have explored new human–robot interaction methods, mostly at the algorithmic level, using gestures, voice and other control methods to optimize the interaction [6–8]. These new interaction methods are more direct and more in line with human thinking habits than the conventional use of interface control and controller control,

providing a new direction for improving human–robot interactions. However, these methods have obvious disadvantages. As the design is focused mainly on the ease of implementation of recognition technology, the real needs of human beings at the interaction level are ignored, resulting in a lack of naturalness in the interaction. For example, some gesture interactions may require the user to make unnatural, complex movements [9], which may increase the user's operational burden.

In some special extreme harsh conditions or movable space limitations and other operator action fatigue operation scenarios, the common form of large-body gesture interaction cannot be applied; in this case, the number of actions is small, and research on the gesture interaction process is not easy. The microgesture interaction scheme can reduce operator fatigue and improve the comfort and efficiency of work. Realizing the control of equipment through tiny finger movements or slight rotation of the wrist is not only flexible in operation but also has very low space requirements; it can also reduce the physical exertion of the operators so that they can maintain good physical condition and work efficiency during long hours of work and bring a more convenient and efficient interaction experience for the actual operation scenario. In this study, user-inspired microgestures for human–robot interaction scenarios are explored to realize more natural and efficient human–robot collaboration.

II. RELATED WORK

A. Microgesture interaction

Gesture interaction is a widely used and ideologically rich interaction method. The growing research on interaction gesture design has greatly increased the usability and sophistication of natural gesture interactions.

In the natural dimension, gestures can be categorized as symbolic, physical, metaphorical, or abstract gestures [10]. In contrast, in this study, we focus on the range of motion of interaction gestures, and in the dimension of range of motion, gestures can include hand movements, arm movements, and

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microgestures. The more common gestures are large limb interaction gestures involving the shoulder, elbow, wrist, etc. This type of aerial gesture often requires the user to raise his/her arm to eye level, which can easily make the user feel fatigued and inefficient [11]. The micromovement gesture is a type of finger-level micromovement that relies on slight knuckle movement to make movements [12], can minimize the movement force of manual action [13], can adapt to the needs of persistent tasks, and, under some maintenance, can detect special applications because of the movable space, machine cost, human physical strength, site noise and other factors of the limitations of the microgesture, is also an alternative to the air gesture, the voice of the available alternative methods of interaction. Compared with arm and hand movements, microgestures travel significantly shorter distances and can therefore quickly complete commands in a short period of time [14]. Existing microgestures can be realized via hardware technologies such as camera-based sensors, pyroelectric infrared sensors, finger-worn computers, millimeter-wave radar, magnetic sensors, and other sensing technologies such as acoustic sensors [15-19], and the required technical equipment is relatively lightweight and convenient.

B. Background on the use of microgestures

In extremely harsh conditions of extreme cold and heat, such as deserts and polar regions, for example, human physiology faces great challenges [20]. In extremely hot deserts, high temperatures cause the body to sweat profusely, leading to dehydration and electrolyte imbalance, which can easily lead to fatigue [21]. In cold polar regions, low temperatures not only consume a large amount of heat to maintain body temperature but also lead to stiffness of the limbs, which greatly reduces the physical strength and reaction ability of human beings [22]. Conventional operating methods often require large body movements, which can accelerate fatigue in extreme environments. For example, frequently lifting the hand to operate a device in a desert area will cause the body to consume more energy and exacerbate dehydration; in polar regions, large movements will cause the body to lose heat more quickly, increasing the risk of hypothermia.

In contrast, microgesture operation has obvious advantages. Micro gestures require only small fine movements of the fingers, consume very little energy, and can effectively reduce the generation of human fatigue. Without increasing the burden on the body, the operator can complete the task in a more durable and stable manner, which greatly improves the efficiency and safety of the operation under extremely harsh conditions.

C. User-inspired design methods for gestures

Gesture design methods can be categorized into two main types: expert-defined methods and user-inspired design methods. Early gesture design was dominated by the expert-defined method, Wobbrock et al. discovered the user's preference for user-induced gestures, and Nielsen et al. proposed a process for extracting and developing user-defined gestures [23-25], which has contributed greatly to gesture design research. The user-inspired design methodology is capable of extracting ergonomic and natural interaction solutions [26], which focuses on the two important goals of reducing the cognitive load and improving the user experience. The general process of user-inspired design

is to define requirements - gesture elicitation - vocabulary definition. In the requirement definition phase, a set of commands is identified; in the gesture elicitation phase, the spontaneous gestures of users triggering the commands are collected via the Wizard of Oz experiment; in the vocabulary definition phase, the gestures are grouped, compared, and analysed on the basis of actions or semantics, e.g., each gesture is ranked according to guessability, and the optimal gesture vocabulary is defined [27-30]. The experimental process can usually be objectively recorded by an RGB camera for gestures. [26-27]. This user-centered gesture design approach is applied in this study to obtain a natural microgesture interaction.

III. EXPERIMENTAL DESIGN

The experiment aims to analyse user-inspired microgesture interaction scenarios corresponding to typical commands in human-robot-machine collaboration scenarios. Using an operator manipulating an intelligent probe vehicle for probing in a desert context as a specific application scenario, a single control command for a typical task was collected to stimulate the microgesture performance of the subjects to study and analyse the microgesture preference of the subjects for a specific task and to formulate a natural microgesture interaction scheme corresponding to the task.

A. Subjects and instruments

A total of 17 subjects participated in the experiment; the subjects were between the ages of 22-30 years, had normal or corrected-to-normal vision, and all had some design foundation or design background. Prior to the experiment, the subjects read the introduction and instructions for the experiment.

We developed the experimental program using the unity engine to present the experiment to the subjects on a large screen connected to a computer, which was connected to the computer to run the program and test it through the unity engine.

B. Experimental design and process

Prior to the experimental design, we conducted a user study on the scenarios of manoeuvring an intelligent rover in a desert situation and collected some typical single tasks, which were focused on two main aspects: azimuthal movement and camera shooting.

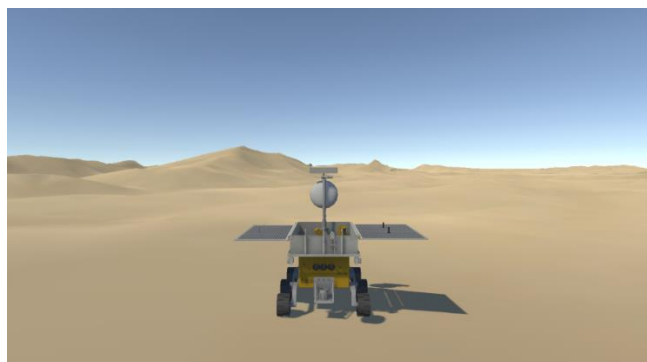


Fig. 1. Experiments in 3D: Intelligent Probe Vehicles in Desert Environments.

The specific single tasks are as follows: (1) moving forward; (2) moving backwards; (3) turning left; (4) turning right; (5)

accelerating; (6) decelerating; (7) unfolding the mini-map; (8) retracting the mini-map; and (9) taking photos.

In the experimental program, we show the subjects the contents of these tasks in the form of 3D scenes in the unity engine and display the name of the task in the upper left corner of the screen. The 3D scene takes the desert as the environment and shows the activities of each task of its intelligent probe vehicle, as shown in Fig. 1. For (9), the experimental screen shows the first view of the camera on the intelligent rover. The others are shown in the regular third view, where the user sees the rover moving.

Before the experiment began, we ensured that the subjects understood the scope of the microgestures that we had set up and asked them to keep their shoulders, elbows, and wrists as still as possible when making microgesture movements and to focus only on the movement of their fingers. Moreover, to stimulate the subjects' ideas about microgestures, we provided prompts in terms of the type of movement of individual fingers, the order of movement of fingers, the use of one/two hands, the use of left/right hands, and the bending/stretching of fingers to inform users that they do not need to limit their ideas but only need to ensure that the movements they make are within the scope of microgestures.

Experiments were conducted via a user-inspired method of collecting gestures, as well as a Wizard of Oz method of responding to the subjects' manipulation gestures [31]. Users were asked to make microgestures in response to activities they saw on the screen, and the master subject would covertly manipulate the program so that once the subject made a microgesture, the master subject would press a key to make the program respond accordingly. The subjects were told that any microgesture they made at the test site would trigger the appropriate function. The participants sat facing a large 2D screen, resting their elbows on the desktop to reduce fatigue and stress, and the unity program was used on the large screen to show the subjects specific tasks in a 3D environment, with the specific scenario being an intelligent probe vehicle in the desert, with a fixed translational distance, a rotational scale, and a zoom scale. Each task was repeated three times: the first two times were used for demonstration, and the third time, the subjects were required to perform the gesture corresponding to that task. The participants were free to choose the one-handed or two-handed microgesture that they thought best suited to the task, encouraging the users to think aloud. There was a 10-s period between each task for subjects to take a break, and the order in which the tasks were presented was randomized.

C. Analysis

In the experiments, videos of the microgestures for each task were collected with cameras at the front and side of the subjects, and the subjects' "thinking out loud" was recorded with an audio recording device during the experiments.

Seventeen participants completed nine tasks, producing a total of 153 gestures. The experiment collected video recordings from cameras with front and side views of each subject, as well as audio recordings of each subject "thinking out loud".

Before deriving the optimal microgesture for each type of gesture from the gesture data, it was necessary to subjectively categorize the gesture results and calculate the degree of consistency of the microgesture design solutions for each task. Referring to the user-inspired design methodology described in previous studies and given the diversity of finger microgestures, gesture classification categorizes variants with common features into one class. We use the Agreement Rate (AR) to calculate the degree of agreement between design solutions via the formula shown below [32]:

$$A = \sum_{P_s \in P_t} \left(\frac{|P_s|}{|P_t|} \right)^2 \quad (1)$$

where A denotes the agreement rate of the microgesture corresponding to a particular task, P_s refers to the number of gestures with microgesture types, and P_t refers to the total number of microgestures. When the agreement rate is greater than 0.3, the agreement rate of the task is relatively high, and when the agreement rate is less than 0.1, the agreement rate of the task is relatively low [33]. In actual processing, although the agreement rate of some instructions is lower than 0.1, it is very close to 0.1. In actual processing, although the agreement rate of some instructions is lower than 0.1, it is very close to 0.1, which is within the acceptable range and can be further analysed and processed by virtue of the number of gesture categories sorted. We calculated the agreement scores of each task according to the formula of the agreement rate, as shown in Fig. 2.

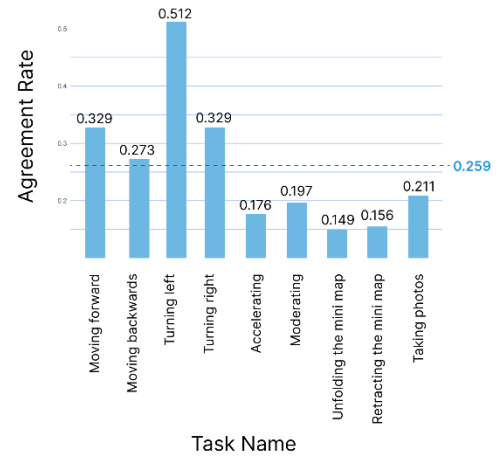


Fig. 2. Agreement rate of microgestures in each task.

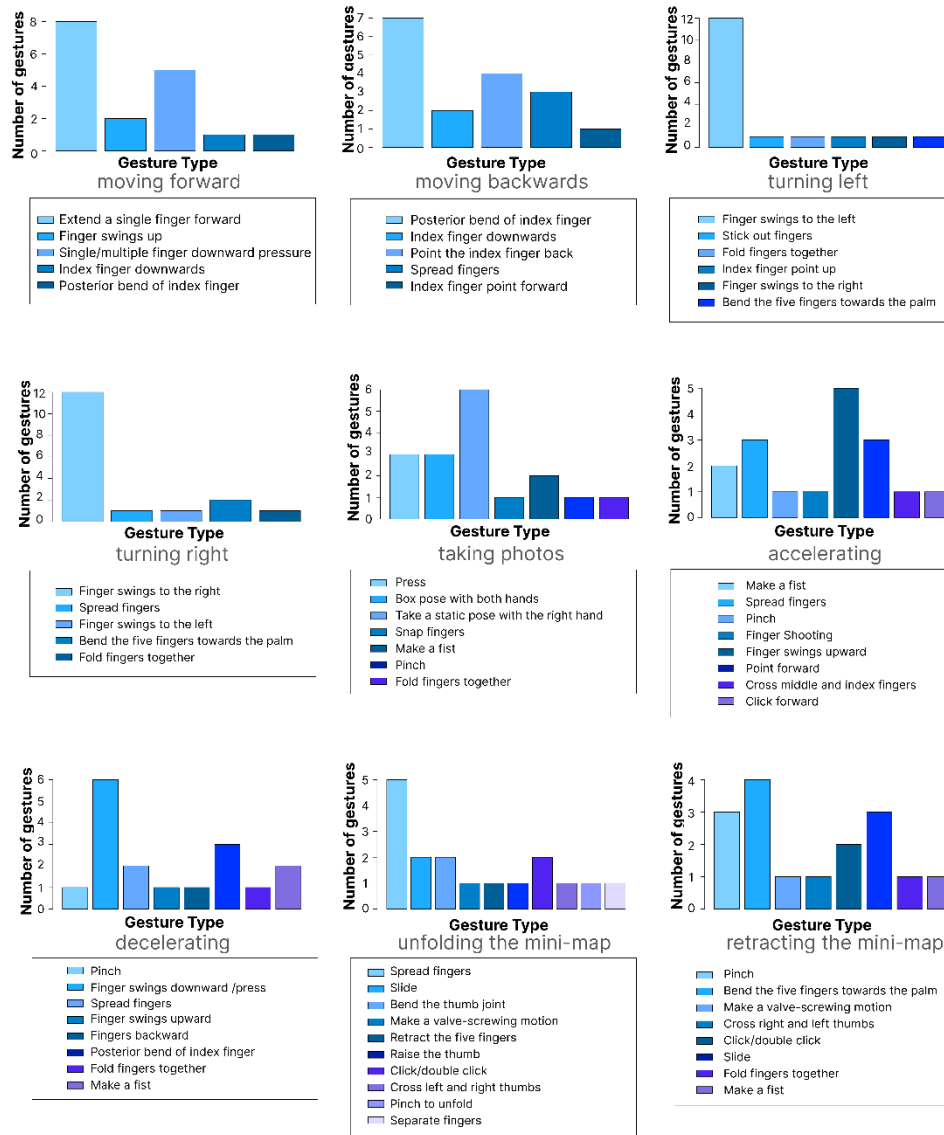


Fig. 3. Hand gesture results collected from the experiment.

The agreement rates for all the tasks are shown in Figure 1. Among these tasks, the agreement rates of backwards, acceleration, deceleration, expanding the mini-map, retracting the mini-map, and taking a picture are low, below 0.3, but greater than 0.1, which can be considered acceptable; however, some tasks have a high agreement rate, and the agreement rate of the forward task, the right-turning line task, is 0.329, which is greater than 0.3, and the agreement rate of the backwards task even exceeds 0.5. The agreement rate is a measure of cognitive load, and when the agreement rate is high, there is a high degree of consistency in the distribution of the microgesture group for a certain task. For microgestures, which are a variation-rich interaction, we lowered the acceptable standard of the agreement rate and considered that all tasks with agreement rates greater than 0.1 have adoptable consistency, which is the basis for us to further extract the best microgestures from the microgesture

groups of a particular task. The average agreement rate of all tasks is 0.259, which is within the range of 0.1--0.3 [34], with a moderately high consistency, indicating that these microgestures are more in line with people's natural interaction habits. The agreement rate of the micro gestures related to orientation movement is higher in this experiment, and the agreement rate of the purely directional commands is higher than that of the acceleration and deceleration commands. The agreement rate of the tasks of forward, backwards, left turn and right turn is 0.360, which is significantly greater than that of the acceleration and deceleration tasks, and the agreement rate of the tasks of acceleration and deceleration is 0.187. The agreement rate is 0.360 for the forward, backwards, left turn, and right turn tasks, which is significantly higher than the average of 0.187 for the acceleration and deceleration tasks, whereas the agreement rates of the map window and photorelated abstraction

tasks are lower. We hypothesize that this is because the task content of azimuthal movement matches naturally with human gesture movement, whereas abstract tasks require people to correspond micro gestures to the concept of the task.

In the specific gesture collection, we found that most of the subjects tended to adopt the index finger for finger micromanipulation movements, and out of the 153 gestures collected, 115 gestures involved the index finger, and 44 adopted the index finger as the only finger to make micromanipulation movements. This may be because the index finger is the usual finger among the five human fingers. This inspires us to consider the use of more habitual fingers for microgesture design in future studies of microgesture interactions. The results of the experimentally collected gestures are shown in Fig. 3.

The user-defined gesture lexical repertoire is determined by ranking the different classes of microneature groups in each task and selecting the largest group of identical gestures to match that particular task. This gesture category metric is called guessability [35], which is used as an indicator of the intuitiveness associated with the gesture rela]guessability is defined as the best microgesture vocabulary for a given task. On the basis of this microgesture vocabulary definition approach, the finalized best microgesture vocabulary for each task is shown in Fig. 4.

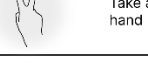
Moving forward		Extend a single finger forward (e.g. index finger forward)
Moving backwards		Posterior bend of index finger
Turning left		Finger swings to the left
Turning right		Finger swings to the right
Accelerating		Finger swings up
Moderating		Finger downward swing/press
Unfolding the mini map		Spread fingers
Retracting the mini map		Bend the five fingers towards the palm
Taking photos		Take a static pose with the right hand (e.g. make a yay/ok pose)

Fig. 4. Best Microgesture Vocabulary for Each Task.

IV. SUMMARIZE

The results of this experimental study show that microgestures have a moderate agreement rate with a mean value of 0.259, making it an interaction scheme with high

usability on the basis of reduced operator hand activity and fatigue. Compared with gesture studies involving large limbs and wrists, the overall agreement between the two groups in Wobbrock et al.'s study was 0.32 and 0.28 [32], and the average agreement rate of the 18 studies calculated by Vatavu et al. was 0.221 [33]. Our results are close to all of them, so in terms of the match and naturalness of semantic expression of gestures, the performance of microgestures deserves to be recognized.

In this study, we analysed the results of user gestures inspired by task content and eventually created a user-defined set of microgestures to match the microgestures that are shown to be dominant in user behavioral data with the corresponding human–robot collaboration commands. Through the study of user-centered microgesture specificity, we have gained more understanding of user thinking and preferences for microgestures in human–robot collaboration scenarios. This microgesture interaction extends the applicability of AR and VR, making gesture interaction possible for special scenarios under spatial and physical constraints.

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